

Math 381 Fourier Analysis and Boundary Value Problems for ISP Notes

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1 Partial Differential Equations

Example 1.1. Find the function $u(x, y)$ such that $u : \mathbb{R}^2 \mapsto \mathbb{R}$ that satisfies the differential equation $u_{xy} = 0$

We can just integrate twice. Once with respect to x and one with respect to y .

$$\begin{aligned}u_{xy} &= 0 \\ \Rightarrow u_x &= c(x) \\ \Rightarrow u(x, y) &= C(x) + D(y)\end{aligned}$$

Example 1.2. Find a solution $u(x, y)$ to solve the transport equation

$$u_x + yu_y = 0$$

Conceptually we can think of this as $\vec{v} \cdot \nabla u = D_{\vec{v}}u = 0$ where $\vec{v} = \langle 1, y \rangle$. We effectively want to travel along a path such that the directional derivative in $\vec{v}$ is 0. These paths are called integral curves as they are tangent to the vector field at every point. We now arrive at the characteristic equation

$$\frac{dy}{dx} = \frac{y}{1}$$

since the slope at every point is just rise over run and therefore $\frac{y}{1}$. This leads to the solution

$$y(x) = Ae^x$$

where A is found using an initial condition. We note that since the value of the function doesn't depend on where you are on each curve but rather *which* curve you're on, we can write $u(x, y) = f(A)$ where $A = ye^{-x}$. Therefore the general solution to the transport equation is

$$u(x, y) = f(ye^{-x})$$